

Annex 1. Detailed Methods for NRF and Ratio Formulas

A1.1 Nutrient Rich Foods Index (NRF_{fish}) Calculation

The overall nutrient density of each species was estimated using a modified Nutrient Rich Foods Index (NRF_{fish}) following Hallström et al. (2019). The index combines qualifying nutrients (those contributing positively to nutritional quality) and limiting nutrients (those to be moderated). Higher NRF_{fish} values indicate a greater nutrient yield per unit of energy. The NRF_{fish} was calculated once using the nutrient values per 100 g and once using the nutrient values per 100 kcal. Values of nutrients per 100 kcal were estimated using the Energy [kcal] values present in the raw data.

$$\text{NRF}_{fish} = \sum_{i=1}^x (\% \text{NRV}_{i, \text{qualifying}}) - \sum_{j=1}^y (\% \text{NRV}_{j, \text{limiting}}) \quad (1)$$

where:

- %NRV = percentage of Nutrient Reference Value per 100 g edible portion; or percentage of Nutrient Reference Value per 100 kcal edible portion
- Qualifying nutrients: Protein, Vitamin A, Vitamin D, Vitamin E, Vitamin C, Vitamin B₁, Vitamin B₂, Vitamin B₃ (Niacin and Niacin Equivalents), Vitamin B₆, Vitamin B₁₂, Folate, Tryptophan, Magnesium, Calcium, Iron, Zinc, and the marine fatty acids EPA+DHA.
- Limiting nutrients: Saturated Fatty Acids (SFA), Cholesterol, and Sodium.

The Nutrient Reference Values (NRVs) were taken from Regulation (EU) No 1169/2011, corresponding to a reference intake of 8400 kJ (2000 kcal) for adults. A reference threshold of 300 mg/day was used for cholesterol for comparability with prior nutrient density models.

A1.2 Ratio Formulas

Derived lipid and mineral ratios were calculated to evaluate the nutritional balance among key nutrients. All ratios are expressed as unitless quotients based on concentrations per 100 g edible portion.

$$\text{PUFA/SFA} = \frac{\text{Polyunsaturated Fatty Acids (PUFA)}}{\text{Saturated Fatty Acids (SFA)}} \quad (2)$$

$$\text{SFA/MUFA} = \frac{\text{Saturated Fatty Acids (SFA)}}{\text{Monounsaturated Fatty Acids (MUFA)}} \quad (3)$$

$$\text{SFA/PUFA} = \frac{\text{Saturated Fatty Acids (SFA)}}{\text{Polyunsaturated Fatty Acids (PUFA)}} \quad (4)$$

$$\text{n-6/n-3} = \frac{\text{Omega-6 Fatty Acids}}{\text{Omega-3 Fatty Acids}} \quad (5)$$

$$\text{Na/K} = \frac{\text{Sodium (Na)}}{\text{Potassium (K)}} \quad (6)$$

$$\text{Ca/P} = \frac{\text{Calcium (Ca)}}{\text{Phosphorus (P)}} \quad (7)$$

$$\text{Fe/Zn} = \frac{\text{Iron (Fe)}}{\text{Zinc (Zn)}} \quad (8)$$

$$\text{EPA/DHA} = \frac{\text{Eicosapentaenoic Acid (EPA)}}{\text{Docosahexaenoic Acid (DHA)}} \quad (9)$$

These indices provide complementary views of dietary quality: lipid ratios reflect the balance between hypocholesterolemic and hypercholesterolemic fatty acids, while mineral ratios capture electrolyte and trace-element equilibrium.

A1.3 Calculation Basis

All nutrient values were derived from raw fish composition per 100 g edible portion using data from the Portuguese Food Composition Table (INSA, 2023). EPA and DHA values were sourced from complementary databases (Norwegian, Dutch, Swedish, and U.S.) and harmonized by fish category (lean vs. oily). All analyses were implemented in Microsoft® Excel® for Microsoft 365 MSO (Version 2509 Build 16.0.19231.20138) 64-bit.

References:

Hallström, E., Bergman, K., Mifflin, K., Parker, R., Tyedmers, P., Troell, M., Ziegler, F. (2019). Combined climate and nutritional performance of seafoods. *Journal of Cleaner Production*, 230, 402-411. <https://doi.org/10.1016/j.jclepro.2019.04.229>

Instituto Nacional de Saúde Doutor Ricardo Jorge (INSA). (2023). Tabela de Composição de Alimentos (TCA) de Portugal Version 2023 Edition). <https://www.insa.min-saude.pt/disponivel-nova-versao-da-tabela-de-composicao-de-alimentos/>

European Food Safety Authority (EFSA). (2010). Scientific Opinion on Dietary Reference Values for fats, including saturated fatty acids, polyunsaturated fatty acids, monounsaturated fatty acids, trans fatty acids, and cholesterol. *EFSA Journal*, 8(3). <https://doi.org/10.2903/j.efsa.2010.1461>